A solution of Erdős Problem 289 with intervals of length two and three

Johan Land
Independent Researcher

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Abstract

Erdős and Graham asked whether, for every sufficiently large integer k, there exist k finite intervals of positive integers, each containing at least two integers and no two of them overlapping or adjacent, such that the reciprocals of all integers in the intervals sum to 1. We prove a stronger statement: for every sufficiently large k the intervals can be chosen to have length two or three and to lie in $[1, 20k]$. The proof combines the Bourgain–Garaev bound for short sums of modular inverses, the structure theorem of Conlon, Fox, He, Mubayi, Pham, Suk and Verstraëte for subset sums, and the Liu–Sawhney density theorem for unit fractions, with a descent through the prime powers of a denominator that cancels one prime power at a time using pairs of consecutive integers while adding a predetermined number of intervals at every stage.

1 Introduction

1.1 The problem and the main theorem

Problem 289 on the Erdős problems site [EP], originating in Erdős and Graham [EG], reads as follows.

> Is it true that, for all sufficiently large k, there exist finite intervals $I_1, \ldots, I_k \subset \mathbb{N}$, distinct, not overlapping or adjacent, with $|I_i| \geq 2$ for $1 \leq i \leq k$, such that
>
> $$1 = \sum_{i=1}^{k} \sum_{n \in I_i} \frac{1}{n}\,?$$

Throughout, an interval of integers $[a, b]$ with $a \leq b$ consists of all integers n with $a \leq n \leq b$, so $|[a, b]| = b - a + 1$. Two intervals $[a, b]$ and $[a', b']$ with $b < a'$ are *adjacent* if $b + 1 = a'$, and *separated* if $b + 1 < a'$, that is, if at least one integer lies strictly between them. A family of intervals is separated if any two of its members are separated after ordering them.

Theorem 1. *For every sufficiently large integer k there are integer intervals $[a_i, b_i]$, $1 \leq i \leq k$, such that*

$$2 \leq a_i \leq b_i \leq 20k, \qquad b_i - a_i + 1 \in \{2, 3\}, \qquad b_i + 1 < a_{i+1} \;\; (1 \leq i < k), \qquad \sum_{i=1}^{k} \sum_{n=a_i}^{b_i} \frac{1}{n} = 1. \quad (1.1)$$

Corollary 1. *The answer to Problem 289 is affirmative: for all sufficiently large k there exist finite intervals $I_1, \ldots, I_k \subset \mathbb{N}$, distinct, not overlapping or adjacent, with $|I_i| \geq 2$, such that $1 = \sum_i \sum_{n \in I_i} 1/n$.*

Proof. Take $I_i = [a_i, b_i]$ from Theorem 1. For $i < j$ the inequalities $b_i + 1 < a_{i+1} \leq a_j$ show that I_i and I_j are disjoint, hence distinct, and that the integer $b_i + 1$ lies in neither of them, so they are not adjacent. Each $|I_i| \in \{2, 3\}$ is at least 2, and the reciprocal sum is the same. □

The separation requirement is the substance of the problem. If adjacent intervals were permitted, any interval of length at least four could be cut into adjacent pieces of lengths two and three, so the number k of intervals would carry no information; and as observed in the discussion of [EP], without the restrictions the question is easy. Theorem 1 says more than the problem asks: the intervals have length two or three, and all of them lie in $[1, 20k]$. The constant 20 is not optimized; it arises from the far band $[4k, 20k]$ used in Section 7, where the only requirement is the inequality $16\gamma > 1$ in (7.2).

1.2 Outline of the proof

The proof adapts the absorption argument in Section 4 of Conlon, Fox, He, Mubayi, Pham, Suk and Verstraëte [CFHMPSV]. Its descending cancellation of denominator prime powers, its small reciprocal budget, and its terminal reservoir form the starting architecture. Lemma 3 below is a sparse version of that paper's inverse-covering theorem, proved by the same bounded-rank progression and divisor-count method. The new ingredients are powersmooth consecutive-pair fibers, the separation of those pairs, a fixed number of pairs at each correction stage, and a core completed by extending pairs to triples.

The final family consists of intervals of three kinds, living at three different scales (Figure 1).

1. **Core pairs** $[Kd+1, Kd+2]$, where K is a fixed integer and d runs over a set $\mathcal{R}$ of parameters of size $\Theta(k^{4/5})$ (Section 6). Some of these pairs will later be extended to the triples $[Kd, Kd+2]$.
2. **Main pairs** $[a, a+1]$ with $3 \mid a$, lying in $[k/1024, k] \cup [4k, 20k]$, all of whose endpoints are $k^{4/5}$-powersmooth (Section 7). There are exactly $k - |\mathcal{R}| - C_H$ of them, where C_H is a number fixed in advance, and they are chosen so that the *initial deficit*
$$r_0 = 1 - (\text{mass of core}) - (\text{mass of main pairs})$$
is a rational number of size about $\delta/2$, with $\delta = 1/1000$, whose denominator has all prime-power divisors at most $H = O(k^{4/5})$.
3. **Correction and auxiliary pairs** at scale between $L^{11/10}$ and $O(k^{22/25})$, added by a descent (Section 5) which cancels the prime powers of the denominator of r_0 one at a time, in decreasing order.

The descent works as follows. Let $q = p^a$ be the largest prime power in the current denominator. Lemma 1 supplies a *fiber* I_q of about q^ε integers m such that $qm + 1$ has only prime-power divisors below $q/2$. The mass of the pair $[qm, qm + 1]$ is
$$w(qm) = \frac{1}{qm} + \frac{1}{qm+1},$$

and modulo q the first term behaves like m^{-1}/q while the second is a junk term whose denominator involves only smaller prime powers. Lemma 3 shows that the inverses of a sparse set of size $q^{7\varepsilon/8}$ already cover every residue modulo q using at most $q^{\varepsilon/2}$ summands. Subtracting the corresponding pair masses therefore removes the factor p^a from the denominator while introducing only smaller prime powers. This is the obstacle peculiar to pairs, as opposed to single unit fractions: the junk term $1/(qm+1)$ must not reintroduce large prime powers, which is exactly what powersmoothness of $qm+1$ guarantees.

To make the total number of intervals predetermined, Lemma 5 reserves, for each prime power q, a family $\mathcal{F}_q$ of exactly $s(q) = \lfloor q^{1/20} \rfloor$ *auxiliary* pairs whose endpoints are $(q/2)$-powersmooth. If the cancellation at q uses $c \le s(q)$ correction pairs, $s(q) - c$ auxiliary pairs are added as well. Every stage then adds exactly $s(q)$ intervals, and the whole descent adds exactly $C_H = \sum_{L<q\le H} s(q)$ intervals. No interval is ever split or merged.

After the descent the residual deficit is j/K for some integer $1 \le j \le j_0$, with j_0 fixed. It is absorbed by extending core pairs to triples: extending $[Kd+1, Kd+2]$ to $[Kd, Kd+2]$ adds $1/(Kd)$, and the Liu–Sawhney theorem (through Lemma 6) supplies, inside each of j disjoint bands of parameters, a set D with $\sum_{d\in D} 1/d = 1$. This adds exactly j/K without changing the number of intervals.

Separation is arranged by construction at each scale: correction pairs start at multiples of four (Lemma 2), auxiliary pairs are chosen at distance at least two from every possible correction endpoint (Lemma 5), core parameters d are only used when the neighbourhood $[Kd-1, Kd+3]$ avoids every correction and auxiliary endpoint, and main pairs lie on a grid of spacing three, far beyond every other endpoint. The final assembly in Section 8 only verifies the count, the mass and the conditions in (1.1). Figure 2 shows how the ingredients depend on one another.

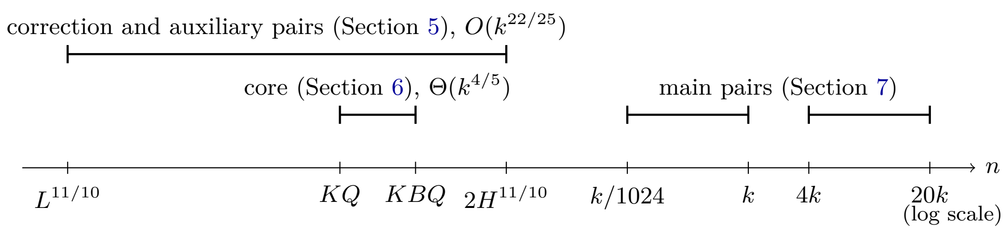

Figure 1: Where the three kinds of intervals live. Here $Q = \lfloor k^{4/5} \rfloor$, $H = KBQ + 2$, and K, B, L are constants.

1.3 Notation and conventions

An integer is *y-powersmooth* if every prime-power divisor of it is at most y. The largest prime-power divisor of an integer always refers to numerical size. For a positive integer a and a modulus m write

$$w(a) = \frac{1}{a} + \frac{1}{a+1}, \qquad e_m(x) = \exp(2\pi i x/m).$$

The *mass* of a set of pairwise disjoint intervals is the sum of the reciprocals of all integers in them; thus the pair $[a, a+1]$ has mass $w(a)$. All logarithms are natural, $\pi(x)$ counts primes up to x, and $\tau(n)$ is the number of divisors of n. A prime power is an integer p^a with p prime and $a \ge 1$; sums and unions over q "prime power" in a range run over all such integers in that range. Asymptotic notation $o(\cdot)$ and $O(\cdot)$ refers to $q \to \infty$ or $k \to \infty$, as indicated, with every constant of Section 2.5 held fixed.

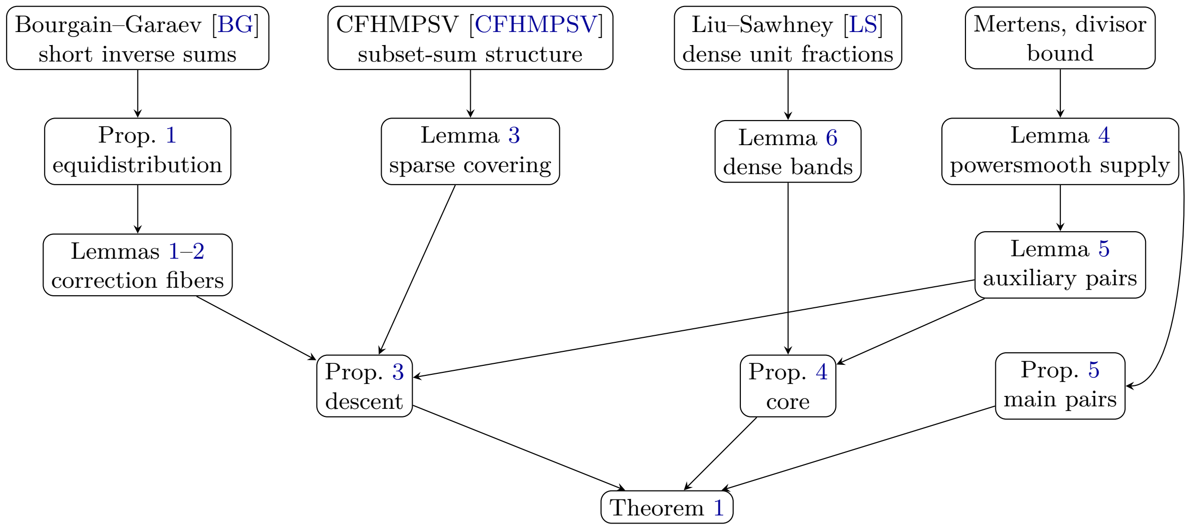

Figure 2: Dependencies among the main statements.

1.4 Formal verification

A formalization of Theorem 1 in Lean 4 with Mathlib is in progress [Repo]. The formal target states (1.1) for intervals indexed by $\{0, \ldots, k-1\}$ with separation required for every ordered pair of indices, exactly as in Theorem 1. We note that the formalization of Problem 289 in the Formal Conjectures library [FC] asks only for disjoint intervals and omits adjacency; Theorem 1 implies both readings.

2 External inputs and choice of parameters

The proof uses three theorems from the literature. Each is stated here in the form in which it is used, together with the place where its hypotheses are verified.

2.1 Short sums of modular inverses

Theorem 2 (Bourgain–Garaev [BG, Theorem 5])**.** *For every sufficiently small fixed $c > 0$, as the integer modulus m tends to infinity, uniformly in $m^c < N < m$ and in coefficients a coprime to m,*

$$\Big|\sum_{\substack{n\le N\\(n,m)=1}} e_m(an^{-1})\Big| = o(N).$$

The quantitative bound in [BG] is $N(\log\log m)^{O_c(1)}/\sqrt{\log m}$. Theorem 2 is used only in the proof of Proposition 1, where it is converted into equidistribution of inverses in intervals by the Erdős–Turán inequality (2.2).

2.2 Structure in bounded subset sums

A generalized arithmetic progression of rank D is a set of the form

$$P = \Big\{ \sum_{i=1}^{D} n_i d_i : \ \alpha_i \le n_i \le \beta_i, \ n_i \in \mathbb{Z} \Big\}$$

with integer generators d_i and real $\alpha_i \le \beta_i$; it is *proper* if distinct coordinate vectors give distinct integers. The dilate tP is defined in Section 4.1.

Theorem 3 (Conlon–Fox–He–Mubayi–Pham–Suk–Verstraëte [CFHMPSV, Theorem 3])**.** *Given fixed $\beta > 1$ and $0 < \eta < 1$, there are constants $c > 0$ and d_0 with the following property. If $A \subseteq [1, n]$, $|A| = m$, $n \le m^\beta$, and*

$$m^\eta \le s \le cm/\log m,$$

then there are $J \subseteq A$, a proper generalized arithmetic progression P of rank at most d_0, and $J' \subseteq J$, such that

$$|J| \ge m - c^{-1} s \log m, \qquad J \cup \{0\} \subseteq P, \qquad |J'| \le s,$$

and the subset sums of J' contain a translate of the proper dilate csP.

This is derived in [CFHMPSV] from Conlon–Fox–Pham [CFP, Theorem 1.5]. We cite the numbering of the journal version of [CFHMPSV], which agrees with arXiv version 2; the numbering differs in the first arXiv version. Theorem 3 is used only in the proof of Lemma 3.

2.3 Dense sets contain a unit-fraction sum of one

Theorem 4 (Liu–Sawhney [LS, Theorem 1.3])**.** *For any fixed $0 < \zeta < 1/2$, every sufficiently large N and every $A \subseteq [1, N]$ with*

$$|A| \ge (1 - 1/e + \zeta)N$$

admit $D \subseteq A$ with $\sum_{d \in D} 1/d = 1$.

Theorem 4 is used only in the proof of Lemma 6.

2.4 Elementary estimates

We use Chebyshev's bound $\pi(x) \ll x/\log x$, Mertens' estimate

$$\sum_{p \le x} \frac{1}{p} = \log\log x + B_1 + o(1) \tag{2.1}$$

for a constant B_1, and the divisor bound $\tau(n) = n^{o(1)}$; see [HW, Theorems 7, 427 and 315]. We also use the Erdős–Turán inequality in the following form [Mo, Chapter 1]: there is an absolute constant C_0 such that for real numbers $x_1, \ldots, x_N$, every interval $J \subseteq [0, 1)$ and every integer $T \ge 1$,

$$\Big| \#\{n \le N : \ \{x_n\} \in J\} - N|J| \Big| \le C_0 \Big(\frac{N}{T} + \sum_{h=1}^{T} \frac{1}{h} \Big| \sum_{n \le N} e(hx_n) \Big| \Big), \tag{2.2}$$

where $\{x\}$ is the fractional part and $e(x) = \exp(2\pi i x)$.

2.5 Choice of parameters

All constants are chosen once, in the following order, and are independent of k.

1. $\varepsilon = 1/10$. Lemmas 1, 2 and 3 are proved for every fixed $\varepsilon \in (0, 1)$; only this value is used afterwards.
2. $\delta = 1/1000$, the approximate size of the initial deficit.
3. An integer L, so large that: Lemmas 1 and 3 (with $\varepsilon = 1/10$) hold for every prime power $q > L$; Lemma 5 holds for the cutoff L; and
$$40L^{-1/20} < \delta/16, \qquad K := \operatorname{lcm}(1, \dots, L) \text{ satisfies } K\delta \geq 2 \text{ and } K \geq 8. \tag{2.3}$$
Each of these conditions holds for all sufficiently large L, so a single choice of L meets all of them.
4. $K = \operatorname{lcm}(1, \dots, L)$, $j_0 = \lceil K\delta \rceil$, and $B = 4^{j_0}$.
5. For every prime power $q > L$, a fiber I_q from Lemma 1 and a family $\mathcal{F}_q$ of $s(q) = \lfloor q^{1/20} \rfloor$ auxiliary pairs from Lemma 5. These determine the endpoint sets $\mathcal{P}^*$, $\mathcal{F}^*$ and $\mathcal{U} = \mathcal{P}^* \cup \mathcal{F}^*$ of Section 5.

The quantities depending on k are then determined in this order:

$$Q = \lfloor k^{4/5} \rfloor, \qquad H = KBQ + 2, \qquad C_H = \sum_{\substack{L < q \leq H \\ q \text{ prime power}}} s(q),$$
$$\mathcal{R} \text{ and } R = |\mathcal{R}| \text{ (Section 6)}, \qquad M = k - R - C_H.$$

The phrase "for sufficiently large k" is used at finitely many places, each time with a threshold depending only on the constants above. The argument yields no explicit value of that threshold, and the constants K and B, as well as the thresholds implicit in Theorems 2–4, are extremely large.

3 Powersmooth fibers and correction pairs

Throughout this section $q = p^a$ is a prime power tending to infinity, $0 < \varepsilon < 1$ is fixed, and $M = q^{\varepsilon}$. A *correction pair* is a pair $[qm, qm + 1]$ with m in the fiber I_q constructed in Lemma 1.

3.1 Equidistribution of inverses in short intervals

We call an integer t *eligible* if t is odd (in case $q = 2^a$) or $(t, 2p) = 1$ (in case $q = p^a$ with p odd). The density of eligible integers is

$$\rho = \tfrac{1}{2} \quad (q = 2^a), \qquad \rho = \tfrac{1}{2}(1 - 1/p) \quad (p \text{ odd}). \tag{3.1}$$

Proposition 1. *Let U_0 be one of the moduli q, $2q$ (if $q = 2^a$) or $4q$, $4pq$ (if p is odd). Every eligible t is a unit modulo U_0, and, as $q \to \infty$, uniformly over all intervals J of integers with $J \subseteq [0, U_0)$,*

$$\#\{t \in [4M, 5M] : \ t \text{ eligible}, \ t^{-1} \bmod U_0 \in J\} = \rho \frac{|J|}{U_0} M + o(M).$$

Proof. Eligibility is precisely the condition $(t, U_0) = 1$ in each case, so the inverse exists. By the Erdős–Turán inequality (2.2), applied to the numbers $x_t = (t^{-1} \bmod U_0)/U_0$ for eligible $t \le N$ with $N \in \{4M, 5M\}$, it suffices to show that for every fixed nonzero integer h,

$$\sum_{\substack{t \le N \\ t \text{ eligible}}} e_{U_0}(ht^{-1}) = o(M) \qquad (q \to \infty), \tag{3.2}$$

since then (2.2) with a fixed T gives discrepancy at most $C_0(N/T) + o(M)$ for the eligible $t \le N$, and T may be taken arbitrarily large after $q \to \infty$. The count over $[4M, 5M]$ is the difference of the counts over $[1, 5M]$ and $[1, 4M]$, with an $O(1)$ error, and the number of eligible $t \le N$ is $\rho N + O(1)$.

To prove (3.2), put $U = U_0/(h, U_0)$ and $h' = h/(h, U_0)$, so that $e_{U_0}(ht^{-1}) = e_U(h't^{-1})$ with $(h', U) = 1$. For large q,

$$q/|h| \le U \le 4q^2,$$

and the prime p still divides U. Fix c with $0 < c < \varepsilon/2$ so small that Theorem 2 applies; then $U^c \le 4^c q^{2c} < M \le N < U$ for large q, so Theorem 2 applies to the modulus U and the lengths $N = 4M$ and $N = 5M$.

If $q = 2^a$, or if p is odd and $2 \mid U$, eligibility is exactly the condition $(t, U) = 1$, and (3.2) is Theorem 2 directly. If p is odd and U is odd, the eligible t are the odd t with $p \nmid t$, that is, the units t modulo U which are odd. Subtract from the sum over all units $t \le N$ the sum over even units. Substituting $t = 2u$ in the latter gives the phase $e_U(h'2^{-1}u^{-1})$, with the unit coefficient $h'2^{-1}$ modulo U, summed over units $u \le N/2$; Theorem 2 applies again to the lengths $2M$ and $5M/2$. Both sums are $o(M)$, which proves (3.2). $\square$

Since the discrepancy bound is absolute, Proposition 1 applies equally to intervals J of relative length $|J|/U_0$ tending to zero, such as the interval of relative length $1/(80p)$ used below; there the main term may be smaller than the error term, and only the resulting lower bound is used.

3.2 Construction of the fibers

Lemma 1 (powersmooth fibers)**.** *Fix $0 < \varepsilon < 1$. For every sufficiently large prime power $q = p^a$ there is a set*

$$I_q \subseteq [q^\varepsilon, 2q^\varepsilon] \cap \mathbb{Z}, \qquad |I_q| \ge q^{\varepsilon - o(1)},$$

such that, for every $m \in I_q$,

$$p \nmid m, \qquad 4 \mid qm, \qquad qm + 1 \text{ is } (q/2)\text{-powersmooth.}$$

The estimate is uniform over prime powers q.

Proof. We construct factorizations

$$qm + 1 = rt, \qquad 4M \le t \le 5M, \qquad \frac{3q}{10} \le r \le \frac{7q}{20}. \tag{3.3}$$

Then $1.2\, qM \le rt \le 1.75\, qM$, so $m = (rt - 1)/q \in [M, 2M]$ for large q, and both $r, t < q/2$. Given t, the condition on r says that the inverse of t modulo an appropriate modulus U_0 lies in a prescribed interval; all counts below are instances of Proposition 1 with the interval $[3q/10, 7q/20]$, of length $q/20 + O(1)$, read modulo U_0.

Case $q = 2^a$. Choose odd t and require the inverse of t modulo q to lie in the interval for r. By Proposition 1 with $U_0 = q$ the number of such t is $M/40 + o(M)$. Among them, $2 \mid m$ if and only if $rt \equiv 1 \pmod{2q}$, that is, if and only if the inverse of t modulo $2q$ lies in the same absolute interval; by Proposition 1 with $U_0 = 2q$ their number is $M/80 + o(M)$. Thus $M/80 + o(M)$ choices have odd m, and $4 \mid qm$ is automatic for large q.

Case p odd. Choose t with $(t, 2p) = 1$ and let r be the inverse of t modulo $4q$, required to lie in the interval in (3.3). Then $qm = rt - 1 \equiv 0 \pmod 4$, so $4 \mid m$. By Proposition 1 with $U_0 = 4q$, the number of such choices is

$$\frac{1}{80} \cdot \frac{1 - 1/p}{2} M + o(M) = \frac{1}{160}(1 - 1/p)M + o(M).$$

Since $4 \mid m$, the additional condition $p \mid m$ is equivalent to $rt \equiv 1 \pmod{4pq}$, that is, to the inverse of t modulo $4pq$ lying in the same absolute interval, which has relative length $1/(80p)$ modulo $4pq$. By Proposition 1 with $U_0 = 4pq$ this happens for $(1 - 1/p)M/(160p) + o(M)$ choices. Subtracting leaves

$$\frac{1}{160}(1 - 1/p)^2 M + o(M) \;\geq\; M/360 + o(M) \tag{3.4}$$

choices of t with $p \nmid m$ and $4 \mid m$.

Powersmoothness. Put

$$\eta = \varepsilon/10000, \qquad D = q^{\varepsilon - \eta}.$$

Remove every t having a prime factor exceeding D. Counting over all $t \leq 5M$ as an upper bound, their number is at most

$$5M \sum_{\substack{D < \ell \leq 5M \\ \ell \text{ prime}}} \frac{1}{\ell} = \Big(5 \log \frac{\varepsilon}{\varepsilon - \eta} + o(1)\Big) M < \frac{M}{1000} \tag{3.5}$$

by (2.1). For a surviving factorization, suppose $u = \ell^b > q/2$ divides rt. Since both factors are less than $q/2$, the prime ℓ divides t; otherwise u would divide r. Hence $\ell \leq D$, and in particular $\ell \neq p$.

There are $O(D \log q)$ prime powers u with $\ell \leq D$ between $q/2$ and $2qM$. For each of them the congruence $qm \equiv -1 \pmod u$ has at most one solution $m \in [M, 2M]$, because $(q, u) = 1$ and $u > q/2 > M$. Each such m arises from at most $\tau(qm + 1) = q^{o(1)}$ factorizations. Deleting them therefore removes at most

$$O(D \log q)\, q^{o(1)} = q^{\varepsilon - \eta + o(1)} = o(M)$$

factorizations. By (3.4), (3.5) and the stronger count in the even case, a positive constant times M factorizations remain, and for each of them $qm + 1$ is $(q/2)$-powersmooth. Dividing by the divisor bound gives $q^{\varepsilon - o(1)}$ distinct valid m. Let I_q be the set of these m. □

3.3 Separation of correction pairs

Lemma 2 (separation of correction pairs). *The pairs $[qm, qm + 1]$, for $q > L$ a prime power and $m \in I_q$, are pairwise separated, including when q varies. More precisely, distinct such pairs have left endpoints differing by at least four.*

Proof. The largest prime-power divisor of qm is q, since $p \nmid m$ and $m < q$. Equal left endpoints therefore force equal q and m. All left endpoints are multiples of four, so distinct ones differ by at least four, leaving at least two unused integers between consecutive pairs. □

4 Sparse modular inverse subsets cover all residues

4.1 Dilates of generalized arithmetic progressions

For a progression $P = \{\sum_i n_i d_i : \ \alpha_i \le n_i \le \beta_i, \ n_i \in \mathbb{Z}\}$ and $t > 0$, the dilate tP used in Theorem 3 expands the coordinate intervals while retaining the generators: it uses integer coordinates $n_i \in [t\alpha_i, t\beta_i]$. Thus its generators, and in rank one its common difference, are unchanged. For noninteger t this depends on the chosen representation of P; throughout, the representation supplied by Theorem 3, and the dilate defined from it, are kept fixed.

4.2 A pigeonhole lemma

Proposition 2 (simultaneous approximation)**.** *Let $q \ge 2$, let $d_1, \dots, d_d$ be integers, and let $a_1, \dots, a_d \ge 1$ be real numbers with $V = \prod_i a_i < q$. Put*

$$B_i = \frac{4q}{a_i}\Big(\frac{V}{q}\Big)^{1/d}.$$

Then there are $1 \le T < q$ and integers $e_i \equiv Td_i \pmod q$ with $|e_i| \le B_i$ for all i.

Proof. Consider the q vectors $(td_i \bmod q)_{1\le i\le d}$, $0 \le t < q$, in $[0, q)^d$. Ignore the coordinates with $B_i \ge q$, and partition each remaining coordinate range $[0, q)$ into intervals of length B_i, at most $2q/B_i$ of them. Note that $\prod_i(q/B_i) = q/4^d$, and that $B_i \le 4q$ for every i because $V < q$ and $a_i \ge 1$. If $r \ge 1$ coordinates remain, the number of boxes is at most

$$\prod_{i:\, B_i<q} \frac{2q}{B_i} \le 2^r \cdot \frac{q}{4^d} \cdot 4^{d-r} = q/2^r < q.$$

If no coordinate remains there is one box. In either case two of the q vectors share a box. Their difference, taken with positive sign, has the form $(Td_i \bmod q)_i$ with $1 \le T < q$, and each coordinate is congruent to an integer of absolute value at most B_i; ignored coordinates impose no restriction because $B_i \ge q$ there. □

The corresponding step in [CFHMPSV] is formulated with a coprimality hypothesis on the generators d_i. That hypothesis is not available in the setting of Lemma 3, where only the elements of J, not the generators of P, are known to be units modulo q; the pigeonhole argument above does not need it.

4.3 The covering lemma

Lemma 3 (sparse inverse covering)**.** *Fix $0 < \varepsilon < 1$. For every sufficiently large prime power q, if*

$$I \subseteq [q^\varepsilon, 2q^\varepsilon] \cap \mathbb{Z}, \qquad (i, q) = 1 \quad (i \in I), \qquad |I| \ge q^{7\varepsilon/8},$$

then every residue modulo q is a sum of inverses of at most $q^{\varepsilon/2}$ distinct members of I.

This is a sparse extension of [CFHMPSV, Theorem 2]. The proof follows their progression and divisor-count argument, with the density loss accounted for explicitly.

Proof. Put $s = \lfloor q^{\varepsilon/2} \rfloor$ and $m = |I|$, and represent the inverses of the members of I by integers in $[1, q-1]$; let A be the set of these representatives. Apply Theorem 3 with $\beta = 2/\varepsilon$, $\eta = 1/3$ and ambient parameter $n = q$. For sufficiently large q,

$$q \le m^{2/\varepsilon}, \qquad m^{1/3} \le s \le cm/\log m,$$

so we obtain $J \subseteq A$, P and J' with $|J| \ge m/2$. Keep the representation defining the theorem's dilate fixed:

$$P = \Big\{ \sum_{i=1}^{D} n_i d_i : \ \alpha_i \le n_i \le \beta_i, \ n_i \in \mathbb{Z} \Big\}, \qquad D \le d_0.$$

Choose an integer coordinate vector v representing $0 \in P$. Put $\ell_i = \lceil \alpha_i \rceil$ and $u_i = \lfloor \beta_i \rfloor$. Call a coordinate *active* when $u_i > \ell_i$; all other coordinates are constant on P. Relabel the active coordinates $1, \dots, d$ and put

$$a_i = u_i - \ell_i \ge 1, \qquad V = \prod_{i=1}^{d} a_i.$$

Every $j \in J$ satisfies

$$j = \sum_{i=1}^{d} (n_i - v_i) d_i, \qquad |n_i - v_i| \le a_i, \tag{4.1}$$

since the contributions of the constant coordinates vanish after subtracting the representation of zero. There is at least one active coordinate, because P contains zero and the nonzero elements of J.

Upper bound for V. The supplied proper progression csP is nonempty. Fix each inactive coordinate at any admissible integer in its original dilated interval. This produces a face of that same proper progression; no representation is changed and no new dilation is formed. For an active coordinate, its dilated interval contains at least

$$cs(\beta_i - \alpha_i) - 1 \ge csa_i - 1 \ge (cs/2)a_i$$

integers for large q. Thus the face has at least $(cs/2)^d V$ points. Its supplied translate lies in the subset sums of J', which lie in $[0, sq]$. Consequently

$$(cs/2)^d V \le sq + 1, \qquad V \ll_\varepsilon qs^{1-d} \ll_\varepsilon q^{1-(d-1)\varepsilon/2}. \tag{4.2}$$

If $V \ge q$ this already forces $d = 1$, and the final paragraph below applies. Assume $V < q$.

Lower bound for V. Apply Proposition 2 to the active generators $d_1, \dots, d_d$ and the numbers a_i, obtaining $1 \le T < q$ and $e_i \equiv Td_i \pmod q$ with $|e_i| \le B_i$. For each $j \in J$, by (4.1) the centered representative h of $Tj \bmod q$ satisfies

$$|h| \le \min\{q/2, R_0\}, \qquad R_0 = \sum_i a_i B_i = 4dq(V/q)^{1/d}.$$

Pair this h with the $i \in I$ for which $j \equiv i^{-1} \pmod q$. There are at least $|J|$ distinct ordered pairs (i, h), even if multiplication by T is not injective on J. They satisfy

$$ih = qz + T, \qquad |z| \le 1 + 8dq^{\varepsilon}(V/q)^{1/d}.$$

The product ih is nonzero because $1 \le T < q$, and $|ih| \le q^{1+\varepsilon}$. For each z the divisor bound allows at most $q^{o(1)}$ ordered pairs (i, h). Therefore

$$\frac{m}{2} \le |J| \le q^{o(1)}(1 + q^{\varepsilon}(V/q)^{1/d}).$$

Since $m \geq q^{7\varepsilon/8}$, the additive 1 is absorbed for large q, giving

$$V \geq q^{1-d\varepsilon/8-o(1)}. \tag{4.3}$$

For $d \geq 2$ this contradicts (4.2), because

$$\frac{(d-1)\varepsilon}{2} - \frac{d\varepsilon}{8} = \frac{(3d-4)\varepsilon}{8} > 0.$$

The rank is bounded independently of q, so these comparisons are uniform. Thus $d = 1$.

Conclusion. By (4.1) every $j \in J$ is a multiple of d_1. Since these j are units modulo q, $(d_1, q) = 1$. The translated one-dimensional face of the original dilate contained in the subset sums of J' has at least $(c/2)sV$ terms. If $V \geq q$ this exceeds q for large q; otherwise (4.3) gives

$$(c/2)sV \geq (c/2)q^{1+3\varepsilon/8-o(1)} > q.$$

An arithmetic progression with common difference coprime to q and at least q terms covers every residue modulo q. Every such term is a subset sum of J', which has at most s members, that is, a sum of at most s inverses of distinct members of I. □

5 The correction and descent mechanism

From now on $\varepsilon = 1/10$, and L, K, δ are as in Section 2.5.

5.1 Powersmooth supply

Lemma 4 (powersmooth supply)**.** *Let $0 < \theta < 1$ and $0 < a < b$ be fixed, let $x \to \infty$, and let $y = x^{\theta+o(1)}$. The number of integers in $[ax, bx]$ that are not y-powersmooth is at most*

$$((b-a)\log(1/\theta) + o(1))x. \tag{5.1}$$

Proof. The number of integers in $[ax, bx]$ divisible by some prime $p > y$ is at most

$$(b-a)x \sum_{y<p\leq bx} \frac{1}{p} + O(\pi(bx)) = ((b-a)\log(1/\theta) + o(1))x$$

by (2.1) and $\pi(bx) = o(x)$. For each prime $p \leq y$ let u_p be the smallest power of p exceeding y. The number of integers at most bx divisible by some u_p is at most

$$\sum_{p\leq y} \left\lfloor \frac{bx}{u_p} \right\rfloor \leq bx\frac{\pi(y)}{y} = o(x).$$

Every integer that is not y-powersmooth is counted by one of the two bounds. □

5.2 Possible correction endpoints

Define the set of *possible correction endpoints*

$$\mathcal{P}^* = \bigcup_{\substack{q>L \\ q \text{ prime power}}} \; \bigcup_{\substack{q^{1/10}\leq m\leq 2q^{1/10} \\ m\in\mathbb{Z}}} \{qm,\ qm+1\}.$$

It contains every endpoint of every correction pair $[qm, qm + 1]$, $m \in I_q$, whichever fibers were chosen. If an element of $\mathcal{P}^*$ is at most X then $q^{11/10} \le qm \le X$, so $q \le X^{10/11}$. Since there are $O(Y/\log Y)$ prime powers up to Y,

$$|\mathcal{P}^* \cap [1, X]| \ll \sum_{\substack{q \le X^{10/11} \\ q \text{ prime power}}} q^{1/10} \ll \frac{X}{\log X}. \tag{5.2}$$

5.3 Auxiliary pairs

Lemma 5 (auxiliary pairs)**.** *For every sufficiently large L one can fix, simultaneously for each prime power $q > L$, a family $\mathcal{F}_q$ of exactly*

$$s(q) = \lfloor q^{1/20} \rfloor$$

pairs $[a, a + 1]$ satisfying

$$q^{11/10} \le a, \qquad a + 1 \le 2q^{11/10}, \qquad 4 \mid a,$$

with both endpoints $(q/2)$-powersmooth, such that all auxiliary pairs, over all $q > L$, are pairwise separated, and every auxiliary endpoint has distance at least two from every element of $\mathcal{P}^*$. In particular every auxiliary pair is separated from every correction pair. If $\mathcal{F}^*$ is the set of all auxiliary endpoints, then

$$|\mathcal{F}^* \cap [1, X]| = O(X^{21/22}). \tag{5.3}$$

Proof. Construct the families in increasing order of the numerical prime power q. Put $x = q^{11/10}$ and $y = q/2$, so $\theta = 10/11$ in Lemma 4. The candidate pairs $[a, a + 1]$ with $4 \mid a$ contained in $[x, 2x]$ number $x/4 + O(1)$; they are pairwise separated. Each integer that is not y-powersmooth spoils at most one candidate, so at least

$$\Big(\frac{1}{4} - \log\frac{11}{10} + o(1)\Big)x \tag{5.4}$$

candidates have both endpoints y-powersmooth; the constant is positive.

Delete every candidate having an endpoint within distance one of $\mathcal{P}^*$. By (5.2) this deletes only $O(x/\log x) = o(x)$ candidates. Delete also every candidate having an endpoint within distance one of a previously chosen auxiliary endpoint. The number deleted in this second step is at most a constant times

$$\sum_{\substack{q' < q \\ q' \text{ prime power}}} s(q') \le q^{21/20} = o(q^{11/10}) = o(x).$$

Thus a positive constant times x candidates remain, more than $s(q)$ for all sufficiently large q. Select any $s(q)$ of them as $\mathcal{F}_q$. All estimates are uniform in q, so a single sufficiently large L suffices for the entire ascending construction. Separation between auxiliary pairs of the same q holds because all candidates are separated; between different q it holds by the second deletion; and every auxiliary endpoint has distance at least two from $\mathcal{P}^*$ by the first deletion, so an auxiliary pair and a correction pair, whose endpoints lie in $\mathcal{P}^*$, are separated.

For (5.3), an auxiliary endpoint at most X has label $q \le X^{10/11}$, so

$$|\mathcal{F}^* \cap [1, X]| \le 2 \sum_{\substack{q \le X^{10/11} \\ q \text{ prime power}}} q^{1/20} = O(X^{21/22}). \qquad \square$$

The families $\mathcal{F}_q$ depend only on L; they are fixed once and for all, independently of k. Put $\mathcal{U} = \mathcal{P}^* \cup \mathcal{F}^*$. By (5.2) and (5.3),

$$|\mathcal{U} \cap [1, X]| = O(X/\log X). \tag{5.5}$$

5.4 The cancellation identity

Remark 1. For a correction pair $[qm, qm+1]$ with $m \in I_q$, in the ring of rationals whose denominators are coprime to p,

$$q\, w(qm) = \frac{1}{m} + \frac{q}{qm+1} \equiv m^{-1} \pmod{q}, \tag{5.6}$$

because $qm+1$ is $(q/2)$-powersmooth and $p \nmid m$. This is the identity behind the descent: subtracting pair masses whose m^{-1} sum to the right residue removes the factor q from the denominator. The proof of Proposition 3 makes this explicit with a common denominator.

5.5 The descent

Proposition 3 (descent). *Let L, K, δ, the fibers I_q and the families $\mathcal{F}_q$ be as in Section 2.5. Let $H > L$ be an integer and let r_0 be a rational number with $r_0 \geq \delta/4$ whose reduced denominator has no prime-power divisor exceeding H. Then there is a set $\mathcal{C}$ of exactly*

$$C_H = \sum_{\substack{L<q\leq H \\ q \text{ prime power}}} s(q) \leq H^{21/20} \tag{5.7}$$

pairs, each either a correction pair $[qm, qm+1]$ with $m \in I_q$ or a member of $\mathcal{F}_q$, for some prime power $q \in (L, H]$, such that:

(i) the pairs in $\mathcal{C}$ are pairwise separated;

(ii) every endpoint of a pair in $\mathcal{C}$ lies in $\mathcal{U}$ and is at most $2H^{11/10} + 1$;

(iii) the mass $W_{\mathcal{C}}$ of $\mathcal{C}$ satisfies $W_{\mathcal{C}} \leq 40L^{-1/20} < \delta/16$;

(iv) $r = r_0 - W_{\mathcal{C}}$ satisfies $r \geq r_0 - \delta/16 > 0$, and its reduced denominator divides K.

Proof. Visit every prime power $q \in (L, H]$ in decreasing numerical order, maintaining a current deficit, initially r_0. The invariant before stage q is that no prime power larger than q divides the current reduced denominator; it holds initially by hypothesis.

Stage $q = p^a$. If q is the full power of p in the current denominator, write the current deficit as $u/(qv)$ with $p \nmid uv$. The fiber I_q satisfies the hypotheses of Lemma 3: its members are coprime to q, and $|I_q| \geq q^{\varepsilon - o(1)} \geq q^{7\varepsilon/8}$ for $q > L$. Lemma 3 therefore selects $S \subseteq I_q$ with $c_q = |S| \leq s(q)$ such that

$$\sum_{m \in S} m^{-1} \equiv uv^{-1} \pmod{q}.$$

Subtract the masses $w(qm)$, $m \in S$, from the deficit. To see the effect, let D be the least common multiple of v and of all the integers m and $qm+1$ for $m \in S$. Then $p \nmid D$, because $p \nmid m$ and $qm+1$ is $(q/2)$-powersmooth, and

$$qD\Big(\frac{u}{qv} - \sum_{m \in S} w(qm)\Big) = \frac{uD}{v} - \sum_{m \in S} \frac{D}{m} - q \sum_{m \in S} \frac{D}{qm+1}$$

is an integer divisible by q, by the chosen congruence. The corrected deficit therefore has the form z/D with $z \in \mathbb{Z}$, so its reduced denominator is coprime to p: the full factor p^a has been eliminated. Every newly introduced prime power is smaller than q, because $m < q$ and $qm + 1$ is $(q/2)$-powersmooth. If the current denominator has no prime-power factor numerically equal to q, set $S = \varnothing$ and $c_q = 0$.

In either case, subtract next the masses of any $s(q) - c_q$ pairs from $\mathcal{F}_q$. Their denominators introduce only prime powers at most $q/2$. They may reintroduce a smaller power of the same prime p, which is handled at its later turn. After the complete stage every prime power in the denominator is strictly smaller than q, which is the invariant for the next stage, including at stages using only auxiliary pairs.

Bookkeeping. Each stage adds exactly $s(q)$ pairs, so $|\mathcal{C}| = C_H$, and $C_H \le \sum_{n \le H} n^{1/20} \le H^{21/20}$. Every selected pair begins at $q^{11/10}$ or beyond, so the pairs of stage q have total mass at most

$$2s(q)q^{-11/10} \le 2q^{-21/20}, \tag{5.8}$$

and their endpoints are at most $2H^{11/10} + 1$. Summing (5.8) over all integers $q > L$,

$$W_{\mathcal{C}} \le 2 \sum_{n > L} n^{-21/20} \le 40L^{-1/20} < \delta/16$$

by (2.3). Correction endpoints lie in $\mathcal{P}^*$ and auxiliary endpoints in $\mathcal{F}^*$, so all endpoints lie in $\mathcal{U}$. Correction pairs are pairwise separated by Lemma 2; auxiliary pairs are pairwise separated and separated from every correction pair by Lemma 5. On completion the reduced denominator has only prime powers at most L, each of which divides $K = \mathrm{lcm}(1, \ldots, L)$; hence the reduced denominator of r divides K, and $r \ge r_0 - W_{\mathcal{C}} > r_0 - \delta/16 \ge \delta/4 - \delta/16 > 0$. □

6 The core

6.1 Dense bands contain a unit-fraction sum

Lemma 6 (dense bands)**.** *If $T \to \infty$ through positive integers and $E_T \subseteq [T, 4T) \cap \mathbb{Z}$ has size $o(T)$, then $[T, 4T) \setminus E_T$ has a subset whose reciprocals sum to 1.*

Proof. The density of $[T, 4T) \setminus E_T$ in $[1, 4T]$ tends to $3/4 > 1 - 1/e + 1/20$. Apply Theorem 4 with $\zeta = 1/20$ and $N = 4T$. □

6.2 Core pairs and their extensions

For $k \to \infty$ put, as in Section 2.5,

$$Q = \lfloor k^{4/5} \rfloor, \qquad H = KBQ + 2,$$

and define the set of *reserved parameters*

$$\mathcal{R} = \{d \in [Q, BQ) \cap \mathbb{Z} : \ [Kd - 1, Kd + 3] \cap \mathcal{U} = \varnothing\}, \qquad R = |\mathcal{R}|. \tag{6.1}$$

The protected interval $[Kd - 1, Kd + 3]$ contains the one-integer neighbourhood of the possible core triple $[Kd, Kd + 2]$. Thus neither that triple nor its shorter pair $[Kd + 1, Kd + 2]$ can overlap or be adjacent to any pair with endpoints in $\mathcal{U}$, in particular to any pair in the set $\mathcal{C}$ of Proposition 3.

Proposition 4 (core)**.** *For sufficiently large k the following hold.*

(i) $R = (B-1)Q - o(Q)$*; in particular* $R = o(k)$*, and* $C_H = O(k^{21/25}) = o(k)$*.*

(ii) The core pairs $[Kd+1, Kd+2]$*,* $d \in \mathcal{R}$*, are pairwise separated, as are the triples* $[Kd, Kd+2]$*,* $d \in \mathcal{R}$*, and every endpoint is at most* H*. Their mass satisfies*

$$W_{\text{core}} \le \frac{2}{K} \sum_{Q \le d < BQ} \frac{1}{d} = \frac{2j_0 \log 4}{K} + o(1) \le 4\delta \log 4 + o(1) < \frac{1}{8}. \tag{6.2}$$

(iii) There are pairwise disjoint sets $D_0, \dots, D_{j_0-1} \subseteq \mathcal{R}$ *with* $\sum_{d \in D_i} 1/d = 1$ *for each* i*. Consequently, for every integer* $1 \le j \le j_0$*, replacing the core pair* $[Kd+1, Kd+2]$ *by the triple* $[Kd, Kd+2]$ *for every* $d \in D_0 \cup \dots \cup D_{j-1}$ *adds exactly* j/K *to the mass and changes neither the number* R *of core intervals nor their separation.*

Proof. (i) By (5.5), only $O_{K,B}(Q/\log Q) = o(Q)$ parameters $d \in [Q, BQ)$ are excluded from $\mathcal{R}$, so $R = (B-1)Q - o(Q)$, which is $\Theta(k^{4/5}) = o(k)$. By (5.7), $C_H \le H^{21/20} = O(Q^{21/20}) = O(k^{21/25})$.

(ii) Distinct triples $[Kd, Kd+2]$ have starts differing by at least $K \ge 8$, so they are separated, and so are the pairs they contain. All endpoints are at most $KBQ + 2 = H$. The mass bound (6.2) uses $\sum_{Q \le d < BQ} 1/d = \log B + o(1) = j_0 \log 4 + o(1)$ and $j_0/K \le \delta + 1/K \le 2\delta$ by (2.3).

(iii) Apply Lemma 6 in the j_0 disjoint bands

$$[4^i Q, 4^{i+1} Q), \qquad 0 \le i < j_0,$$

all of which lie inside $[Q, BQ)$ because $B = 4^{j_0}$. Each band loses only the $o(Q) = o(4^i Q)$ parameters excluded from $\mathcal{R}$, and j_0 is fixed, so Lemma 6 with $T = 4^i Q$ gives $D_i \subseteq \mathcal{R} \cap [4^i Q, 4^{i+1} Q)$ with $\sum_{d \in D_i} 1/d = 1$. The sets D_i are disjoint because the bands are. Extending $[Kd+1, Kd+2]$ to $[Kd, Kd+2]$ adds $1/(Kd)$ to the mass, so extending over $D_0 \cup \dots \cup D_{j-1}$ adds exactly j/K. The extended family is still separated by (ii), and its cardinality is still R. □

7 Main pairs and the exact interval count

By Lemma 4 with $y = Q = k^{4/5+o(1)}$ and $\theta = 4/5$, in any fixed interval $[ak, bk]$ at most

$$((b-a)\log(5/4) + o(1))k$$

integers are not Q-powersmooth. Consider the *grid pairs* $[a, a+1]$ with $3 \mid a$. Distinct grid pairs are separated, and an integer that is not Q-powersmooth spoils at most one of them. Thus $[ak, bk]$ contains at least

$$\gamma(b-a)k + o(k) \text{ grid pairs with both endpoints } Q\text{-powersmooth}, \qquad \gamma = \frac{1}{3} - \log\frac{5}{4} > 0, \tag{7.1}$$

where only $O(1)$ pairs are affected by the boundary. Numerically $\gamma = 0.110\dots$, so

$$10\gamma > 1, \qquad 16\gamma > 1. \tag{7.2}$$

Proposition 5 (main pairs)**.** *Let $M = k - R - C_H$ and $\tau_k = 1 - W_{\mathrm{core}} - \delta/2$. For sufficiently large k there is a set $\mathcal{M}$ of exactly M grid pairs with both endpoints Q-powersmooth, contained in $[k/1024, k] \cup [4k, 20k]$, pairwise separated, whose mass W_{main} satisfies*

$$\tau_k - 2048/k \le W_{\mathrm{main}} \le \tau_k. \tag{7.3}$$

Consequently the initial deficit

$$r_0 = 1 - W_{\mathrm{core}} - W_{\mathrm{main}} \in [\delta/2,\ \delta/2 + 2048/k] \tag{7.4}$$

is a rational number whose reduced denominator has no prime-power divisor exceeding H.

Proof. By Proposition 4(i), $M = k - o(k)$ is a positive integer for large k. Let $\mathcal{A}$ be the set of all grid pairs with both endpoints Q-powersmooth contained in $[k/1024, k]$. Divide this interval into the ten dyadic bands $[t, 2t]$, $t = 2^{-i}k$, $1 \le i \le 10$. By (7.1) each band contributes at least $\gamma t + o(k)$ pairs, each of mass at least $1/t$, so

$$\sum_{A\in\mathcal{A}} \sum_{n\in A} \frac{1}{n} \ge 10\gamma + o(1) > 1 \tag{7.5}$$

by (7.2). There are fewer than $k/3$ pairs in $\mathcal{A}$, hence fewer than M.

The far band $[4k, 20k]$ contains more than k grid pairs with both endpoints Q-powersmooth, by (7.1) and $16\gamma > 1$. Choose any M of them. Their total mass is at most $M/(2k) < 1/2$.

Now replace far pairs, one at a time, by the pairs of $\mathcal{A}$, until every pair of $\mathcal{A}$ has been inserted. Each replacement removes a pair of mass less than $1/(2k)$ and inserts one of mass at most $2048/k$ and at least $2/(k+1)$, so it increases the mass by at most $2048/k$. By (7.5) the mass at the end exceeds 1. Since $W_{\mathrm{core}} < 1/8$ by (6.2), the target τ_k lies in $(1/2, 1)$ for large k. Stop at the last configuration before the mass first exceeds τ_k; this gives (7.3). The count is exactly M throughout. The pairs are separated within each band by the grid, and the two bands are separated from one another.

Finally, (7.4) follows from (7.3). Every main endpoint is Q-powersmooth, and every core endpoint is at most H by Proposition 4(ii), so all prime powers in the reduced denominator of r_0 are at most H. $\square$

8 Proof of the main theorem

Let k be sufficiently large for Propositions 4 and 5 and for the estimates below. Construct, in this order:

1. the core pairs $[Kd + 1, Kd + 2]$, $d \in \mathcal{R}$, and the sets $D_0, \dots, D_{j_0-1}$ of Proposition 4;
2. the $M = k - R - C_H$ main pairs $\mathcal{M}$ of Proposition 5, with the initial deficit r_0 of (7.4);
3. the C_H correction and auxiliary pairs $\mathcal{C}$ of Proposition 3, applied to r_0 and $H = KBQ + 2$. The hypotheses hold: $H > L$ for large k, $r_0 \ge \delta/2 \ge \delta/4$, and the denominator of r_0 has prime powers at most H by Proposition 5.

The residual $r = r_0 - W_{\mathcal{C}}$ satisfies, by Proposition 3(iii)–(iv) and (7.4),

$$\frac{\delta}{4} < \frac{\delta}{2} - \frac{\delta}{16} \le r \le \frac{\delta}{2} + \frac{2048}{k} \le \delta$$

for large k, and its reduced denominator divides K. Hence $r = j/K$ with an integer $1 \le j \le K\delta \le j_0$. Extend the core pairs indexed by $D_0 \cup \cdots \cup D_{j-1}$ to triples, as in Proposition 4(iii). The total mass is now

$$W_{\text{core}} + W_{\text{main}} + W_{\mathcal{C}} + \frac{j}{K} = (1 - r_0) + W_{\mathcal{C}} + r = 1.$$

Count. The family consists of R core intervals, M main pairs and C_H pairs from $\mathcal{C}$; no interval was split or merged, so the total number is $R + M + C_H = k$. Every interval has two or three members.

Separation. The pairs of $\mathcal{C}$ are pairwise separated by Proposition 3(i), and their endpoints lie in $\mathcal{U}$, so by the definition (6.1) of $\mathcal{R}$ every core interval, extended or not, is separated from every pair of $\mathcal{C}$. Distinct core intervals are separated by Proposition 4(ii). Main pairs are separated from one another by Proposition 5. Finally, all endpoints of $\mathcal{C}$ are at most

$$2H^{11/10} + 1 = O(k^{22/25}) = o(k),$$

and all core endpoints are at most $H = O(k^{4/5}) = o(k)$, so for large k all of them lie below $k/1024 - 2$, separating them from every main pair. Thus the whole family is separated.

Endpoints. Main endpoints are at most $20k$ and all other endpoints are $o(k)$, so $b_i \le 20k$. Every left endpoint is at least $\min\{L^{11/10}, KQ, k/1024\} \ge 2$.

Listing the intervals in increasing order gives exactly the conditions (1.1). This proves Theorem 1. □

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